

Constructing Inequalities

These are ordered in (roughly) ascending difficulty.

Problem 1

Show that $\mathbb{E}[\ln(X)] \leq \ln(\mathbb{E}[X])$ for any discrete non-negative random variable X .

Solution 1

We shall prove this result true through an application of Reverse Jensen's Inequality.

Note that the function $g(x) = \ln(x)$ is concave as for any

$$\begin{aligned} 0 \leq x_1 < x_2 < \infty, \quad 0 < t < 1, \\ tg(x_1) + (1-t)g(x_2) &= t \ln(x_1) + (1-t) \ln(x_2) = \ln(x_1^t x_2^{1-t}) \\ &\leq \ln(tx_1 + (1-t)x_2) = g(tx_1 + (1-t)x_2) \end{aligned}$$

due to $\ln(x_1^t x_2^{1-t}) - \ln(tx_1 + (1-t)x_2) \leq 0$ due to the AM-GM inequality which we will demonstrate below.

Since $0 < t < 1$, $t = p/q$ for $q > p, p \in \mathbb{Z} \geq 0$ so $1-t = (1-p)/q$, by the weighted AM-GM inequality we have

$$\frac{px_1 + (1-p)x_2}{q} \geq (x_1^p x_2^{1-p})^{1/q}.$$

Simplifying this,

$$tx_1 + (1-t)x_2 \geq x_1^t x_2^{1-t},$$

causing $\ln(tx_1 + (1-t)x_2) \geq \ln(x_1^t x_2^{1-t})$ as log is an increasing function due to $\frac{d}{dx}(\ln(x)) = 1/x$ being non-negative in its domain of $x \in \mathbb{R}^+$.

Letting $f(x)$ be the pmf of X , we know that $\mathbb{E}[X] = \sum x f(x)$ is a sum of discrete values where $f(x) = P(X = x)$ which can be expressed as a fraction over a particular integer n . This means $\mathbb{E}[X] = \sum_{i=0}^n \frac{a_i}{n}$ for a particular n .

Hence by Reverse Jensen's Inequality as $\ln(x)$ is a concave function

$$\begin{aligned} \mathbb{E}[\ln(x)] &= \sum \ln(x) f(x) = \sum \ln(x) b_x / n = 1/n \sum \ln(x^{b_x}) \\ &= 1/n \sum \ln(a_x) \leq \ln((\sum a_x)/n) \\ &= \ln(\sum (x^{b_x})/n) \leq \ln(\sum x b_x / n) = \ln(\sum x f(x)) = \ln(\mathbb{E}[X]). \\ &\quad (x^{b_x} \leq x b_x \text{ as } b_x \text{ is quantised } \mathbb{N}, 0 < x < 1) \end{aligned}$$

Problem 2

Prove that probability of getting doubles when rolling a pair of (identical, but possibly unfair) n -sided die is at least $1/n$.

Solution 2

This problem will involve the use of Jensen's inequality. Let the probability of getting a particular score be p_i , hence doubles as p_i^2 for $i = 1$ to n .

Consider $\frac{P(\text{Doubles})}{n} = \sum_{i=1}^n \frac{p_i^2}{n}$. If we apply Jensen's Inequality to this, noting that x^2 is a convex function since $\frac{d^2}{dx}x^2 = \frac{d}{dx}2x = 2 > 0$, $\frac{P(\text{Doubles})}{n} \geq (\sum_{i=1}^n \frac{p_i}{n})^2$.

This can be further simplified

$$\begin{aligned}\frac{P(\text{Doubles})}{n} &\geq \left(\sum_{i=1}^n \frac{p_i}{n}\right)^2 \\ &= \frac{1}{n^2} \left(\sum_{i=1}^n p_i\right)^2 \\ &= \frac{1}{n^2} \cdot 1\end{aligned}$$

Hence $P(\text{Doubles}) \geq \frac{1}{n}$. Note we can also use Lagrange Multipliers or Cauchy Schwartz on $(p_1, \dots, p_n)^T$, $(1, \dots, 1)^T$ n -dimension vectors.

Problem 3

Let a, b, c be positive real numbers such that $abc = 1$. Prove that:

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}$$

Solution 3

(IMO 1995, Problem 2) First let's use the ratio sub we just learnt about. Let $x = \frac{1}{a}$, $y = \frac{1}{b}$, and $z = \frac{1}{c}$. Because $abc = 1$, it immediately follows that $xyz = \frac{1}{abc} = 1$. Rewriting the first term of our inequality in terms of x, y , and z ,

$$\frac{1}{a^3(b+c)} = \frac{x^3}{\frac{1}{y} + \frac{1}{z}} = \frac{x^3yz}{y+z} = \frac{x^2}{y+z}.$$

So the inequality is therefore equivalent to

$$\frac{x^2}{y+z} + \frac{y^2}{z+x} + \frac{z^2}{x+y} \geq \frac{3}{2}.$$

With the Engel's Form of Cauchy Schwartz,

$$\begin{aligned}
\frac{x^2}{y+z} + \frac{y^2}{z+x} + \frac{z^2}{x+y} &\geq \frac{(x+y+z)^2}{(y+z) + (z+x) + (x+y)} \\
&= \frac{(x+y+z)^2}{2(x+y+z)} \\
&= \frac{x+y+z}{2} \\
&\geq \frac{3\sqrt[3]{xyz}}{2} \quad \text{by AM-GM} \\
&= \frac{3}{2}.
\end{aligned}$$

Problem 4

Let $n \geq 3$ be an integer and $a_2, a_3, \dots, a_n$ positive real numbers such that $a_2 a_3 \dots a_n = 1$. Prove that

$$(a_2 + 1)^2 (a_3 + 1)^3 \dots (a_n + 1)^n > n^n$$

Solution 4

Note that for all $k \geq 2$, we find that

$$\frac{a_k + 1}{k} = \frac{a_k + \overbrace{\frac{1}{k-1} + \frac{1}{k-1} + \dots + \frac{1}{k-1}}^{k-1 \text{ times}}}{k} \geq \sqrt[k]{a_k \cdot \left(\frac{1}{k-1}\right)^{k-1}}$$

from AM-GM, with equality iff $a_k = \frac{1}{k-1}$. Thus, we get that

$$\begin{aligned}
(a_k + 1)^k &= k^k \cdot \left(\frac{a_k + 1}{k}\right)^k \\
&\geq k^k \cdot \left(\sqrt[k]{a_k \cdot \left(\frac{1}{k-1}\right)^{k-1}}\right)^k \\
&\geq k^k \cdot a_k \cdot (k-1)^{-(k-1)}
\end{aligned}$$

for all $k \geq 2$. Substituting this into the LHS gives us

$$\begin{aligned}
(a_2 + 1)^2 (a_3 + 1)^3 \dots (a_n + 1)^n &= \prod_{i=2}^n (a_i + 1)^i \\
&\geq \prod_{i=2}^n i^i \cdot a_i \cdot (i-1)^{-(i-1)} \\
&= a_2 a_3 \dots a_n \cdot n^n \cdot 1^{-1} \\
&= n^n
\end{aligned}$$

Note that equality is only achievable iff $a_2 = 1$, $a_3 = \frac{1}{2}$, $\dots$, $a_n = \frac{1}{n-1}$, however this yields $a_2 a_3 \dots a_n = \frac{1}{(n-1)!} \neq 1$ for $n \geq 3$. Thus, the equality case is never achieved, and $(a_2 + 1)^2 (a_3 + 1)^3 \dots (a_n + 1)^n > n^n$ as desired.

Problem 5

Let $a, b, c, d > 0$ be positive real numbers such that $abcd = 1$ and $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} < a + b + c + d$. Prove that $\frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d} > a + b + c + d$.

Solution 5

From AM-GM we find,

$$\begin{aligned} \frac{1}{4} \left(\frac{a}{b} + \frac{a}{b} + \frac{b}{c} + \frac{a}{d} \right) &\geq \sqrt[4]{\frac{a^3 b}{b^2 c d}} \\ &= \sqrt[4]{a^4} \\ &= a . \end{aligned}$$

Thus, we get

$$\begin{aligned} \frac{1}{4} \left(\sum_{\text{cyc}} \frac{a}{b} + \frac{a}{b} + \frac{b}{c} + \frac{a}{d} \right) &\geq \sum_{\text{cyc}} a \\ \sum_{\text{cyc}} \frac{3}{4} \cdot \frac{a}{b} + \sum_{\text{cyc}} \frac{1}{4} \cdot \frac{a}{d} &\geq \sum_{\text{cyc}} a . \end{aligned}$$

Similarly, we find

$$\begin{aligned} \frac{1}{4} \left(\frac{a}{d} + \frac{a}{d} + \frac{d}{c} + \frac{a}{b} \right) &\geq \sqrt[4]{\frac{a^3 d}{b c d^2}} \\ &= \sqrt[4]{\frac{a^4}{abcd}} \\ &= a \end{aligned}$$

and hence

$$\begin{aligned} \frac{1}{4} \left(\sum_{\text{cyc}} \frac{a}{d} + \frac{a}{d} + \frac{d}{c} + \frac{a}{b} \right) &\geq \sum_{\text{cyc}} a \\ \sum_{\text{cyc}} \frac{1}{4} \cdot \frac{a}{b} + \sum_{\text{cyc}} \frac{3}{4} \cdot \frac{a}{d} &\geq \sum_{\text{cyc}} a . \end{aligned}$$

Adding both of these together yields

$$\sum_{\text{cyc}} \frac{a}{b} + \sum_{\text{cyc}} \frac{a}{d} \geq 2 \sum_{\text{cyc}} a$$

or

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} + \frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d} \geq 2(a + b + c + d) .$$

Now, as $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} < a + b + c + d$, we obtain

$$\begin{aligned} \frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d} &\geq 2(a + b + c + d) - \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \\ &> 2(a + b + c + d) - (a + b + c + d) \\ &= (a + b + c + d) \end{aligned}$$

as desired.